

Final MVP Crop Catalog Fields & Location Comparisons

A practical schema for comparing sourced crop requirements with location-specific temperature, soil, water, region, and planting evidence across Texas.

How the evidence is divided

Dimension	Stored in crop catalog	Compared with
Temperature	Thermal ranges, thresholds, stages, and frost sensitivity	FortyGuard recent/local heat plus NASA long-term climate evidence
Soil	Texture, pH, drainage, and crop effective root-zone range	USDA SSURGO map-unit horizons and restrictions, with farmer/lab overrides
Water	Demand, drought tolerance, and irrigation dependency	NASA precipitation, crop-accessible SSURGO storage, and farmer irrigation status
Region and season	Texas applicability, planting window, and maturity	Farm coordinates, region mapping, planting date, and frost-free evidence
Evidence quality	Sources, confidence, review date, assumptions, and status	Validator inputs; not agronomic conditions

Scope note: SSURGO estimates mapped storage and soil restrictions. It does not report current moisture, rainfall, irrigation access, pumping capacity, or canal allocation.

Comparison model

CropSage compares stable catalog requirements with normalized location evidence. Crop root-depth ranges cap the SSURGO horizon calculation; accessible storage remains separate from seasonal water supply.

Complete Catalog Field Matrix

Group	Crop catalog field	Location evidence	Comparison or purpose
Identity	crop_id	None	Catalog lookup and audit key.
Identity	common_name	None	Displayed in rankings and explanations.
Identity	scientific_name	None	Prevents common-name ambiguity.
Identity	production_use	Optional farmer goal	Separates profiles such as grain corn and silage corn.
Identity	variety_scope	Selected crop type and local cultivar	Flags assumptions before scoring.
Region and season	supported_texas_regions	Farm coordinate mapped to region	Determines whether the profile is supported at the location.
Region and season	regional_suitability	Resolved Texas region	Provides regional prior and explanation.
Region and season	planting_windows_by_region	Region and proposed planting date	Checks whether the date is inside a supported window.
Region and season	days_to_maturity	Planting date, frost-free window, and horizon	Checks whether the crop can mature in the available season.
Temperature	optimal_temperature_range	FortyGuard and NASA temperature evidence	Measures temperature fit.
Temperature	heat_stress_threshold	Stage-aware FortyGuard heat evidence	Creates sourced heat warnings or soft penalties.
Temperature	heat_exposure_duration	FortyGuard threshold-exceedance duration	Controls whether a heat event may create a penalty.
Temperature	heat_sensitive_stages	Planting date and estimated crop stage	Applies heat evidence at the relevant stage.
Temperature	frost_sensitivity	Forecast minimum and frost-free-season evidence	Flags establishment or maturity frost risk.
Temperature	temperature_unit	Normalized provider units	Prevents conversion ambiguity.
Temperature	temperature_measurement_basis	Provider parameter metadata	Prevents comparison of unlike measurements.
Soil	preferred_soil_textures	SSURGO surface texture or farmer/lab override	Produces a soft texture-fit result.
Soil	ph_tolerable_range	SSURGO 0-30 cm mapped pH or laboratory override	Checks whether mapped or measured pH is within range.
Soil	effective_root_zone_depth_cm	SSURGO restriction depth and horizon coverage	Caps horizon water integration and produces an accessible-storage range.
Soil	drainage_requirement	SSURGO drainage class or farmer observation	Flags likely drainage mismatch.
Water	water_demand	Rainfall, climate, irrigation, and crop-accessible soil storage	Compares demand with water evidence without treating storage as supply.

Group	Crop catalog field	Location evidence	Comparison or purpose
Water	drought_tolerance	Rainfall, current moisture, irrigation, and accessible storage	Adds crop resilience context.
Water	irrigation_requirement	Farmer irrigation availability and reliability	Filters or warns when required infrastructure is unavailable.
Evidence	source_ids	None	Allows evidence tracing and validation.
Evidence	confidence	Evidence coverage and runtime conflicts	Contributes to recommendation confidence, not suitability.
Evidence	last_reviewed	Current date and review policy	Flags stale records.
Evidence	assumptions	Farm profile and runtime evidence	Exposes limitations and possible conflicts.
Evidence	catalog_version	Saved recommendation version	Makes recommendations reproducible.
Evidence	record_status	Release policy	Prevents retired or unreviewed records from silent production use.

Root-zone comparison rule

1	Catalog	Published crop min-max effective root-zone depth.
2	SSURGO	Shallowest mapped component restriction; 150 cm fallback when no restriction is identified.
3	Usable depth	Minimum of each crop bound and the mapped/farmer soil limit.
4	Water storage	Integrate horizon awc_r to both usable bounds and report the resulting millimetre range.
5	Guardrail	Do not treat storage as current moisture, rainfall, irrigation, or seasonal sufficiency.

Root-depth sources: FAO-56 Table 22 (SRC-028) for all catalog categories; Texas A&M; AgriLife E-618 (SRC-030) provides regional context for selected crops.